ParaThink: Speech Frontend, Parallel Thinking and Expression

Continuous observation / Shared context / One playback owner

Incoming user audio

Speech Frontend: Native SoulX Control + Independent X2 Text

Independent X2 Streaming ASR
Two native 80 ms audio steps
LISTEN / DECODE to real EOS

Same continuous PCM
Independent text / control

Native SoulX + SenseVoiceSmall
Internal ASR / rules / feedback / reset
Five native user states

Each 160 ms: finish both lanes, atomically publish, then advance. WAIT also completes a step.

Future CoT input

One atomic 160 ms commit: text + states + progress

Shared Context: Available Dialogue + Runtime State

Current EOS-closed text + five states + real progress / One joint batch snapshot / Layer 2 CoT
Backchannel timing suggestions / Layer 1 output / Listen–Speak state
Interruption events / generated, submitted, canceled and played records

Read / Timing

Layer 0
Backchannel Timing
System backchannel
Timing prediction

Read / Output

Layer 1
Expression
Read the joint snapshot
Reply after committed complete
Request thinking

Request

Read / CoT

Layer 2
Async Thinking
Publish CoT only
No turn-action advice
Update one active task

Frontend observation, expression and thinking can run concurrently.

Actual Listen / Speak state + interruption events

System backchannel

Speech text

Text FIFO

Submitted speech text / system backchannel, in order
Chunks carry output_id + chunk_id; accept active output only

Consume in order

Committed nonidle
Match output ID
Stop once

Output Runtime: Streaming TTS + Player

Single arbitration point / single owner of audible output
Priority stop: halt audio + cancel pending output
Reject late text / audio carrying an invalid output_id

Audio heard by the user: never revoked

On a valid stop: execution in the same output runtime

1. **Close old output_id**
Block stale submissions

2. **Halt + cancel**
Cancel queued / buffered output
Discard late text / audio

3. **Closure acknowledged**
Publish Listen + interruption
Keep generated history

4. **New Layer 1 reply**
Read committed complete + text
Fresh output_id; new text

Five states share one commit boundary. Committed valid complete can be used with the current text.

NTF-020 retired: no extra handoff wait. Real-model validation, downstream and CoT integration remain pending.